Adaptive Quantum Error Mitigation Under Fixed Shot Budgets: Operating Regimes, Abstention, and Noise-Model Shift

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Abstract

Quantum error mitigation (QEM) trades bias reduction against auxiliary executions and sampling variance, so a method that helps one circuit may waste resources or become unstable elsewhere. We compare unmitigated estimation, full-assignment-matrix readout mitigation, and linear zero-noise extrapolation (ZNE) under identical total shot budgets. The simulator benchmark combines GHZ and fixed-seed hardware-efficient circuits with widths 2, 4, and 6, depths 1, 3, and 6 where applicable, four main noise levels, three shot budgets, and 50 deterministic repeats. A cost-aware oracle labels 144 operating cells as 103 practical ties, 18 readout winners, 14 ZNE winners, and 9 inconclusive. A frozen, non-ML rule selector reaches only 41.4% decision compatibility on a 29-cell deterministic holdout and 50.0% under four unseen synthetic noise shifts. The benchmark retains 99 explicit readout failures and 179 of 432 method summaries with negative RMSE gain. An additional preregistered ZNE validity gate accepts only 5 of 16 simulated circuit–noise–budget cells; none of the 8 tested hardware-efficient cells passes every check. In a separate factorial study, combined readout mitigation and ZNE has the lowest expectation RMSE in 12 of 16 preregistered conditions, yet all 16 combined-ZNE conditions fail at least one frozen validity criterion. Its near-tie decision-error probability remains between 0.26 and 0.48. These simulated results support selective mitigation and abstention as research questions, but do not validate a general selector or establish hardware performance.

1 Introduction

QEM can reduce expectation-value bias without logical encoding, but its additional circuits and statistical amplification consume finite resources [6, 4, 2]. Comparing methods at unequal execution cost can therefore manufacture an apparent advantage. Even under a fixed budget, the best method can change with circuit structure, observable, noise, and sampling regime.

This work asks when mitigation improves RMSE at fixed total shots, when methods are practically indistinguishable, whether a simple selector can choose safely without the exact reference, when apparent ZNE improvement is credible, and whether readout correction and ZNE cooperate under one resource ledger. The contribution is a reproducible simulator benchmark with repeat-level data, explicit execution ledgers, cost-aware labels, abstention rules, and evaluation under noise-model mismatch. It is not a proposal for new QEM theory or a claim of quantum-hardware advantage.

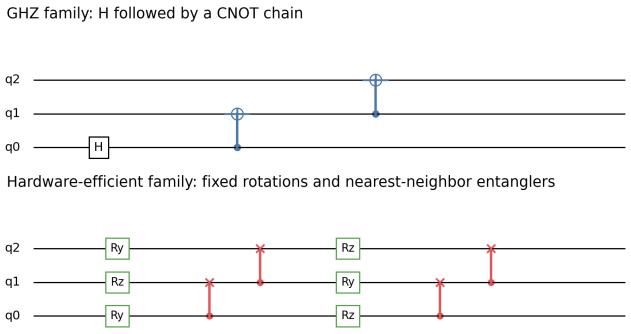


Figure 1: The benchmark combines an analytically controlled GHZ family with a nontrivial hardware-efficient depth grid; both diagrams are conceptual and all executed circuits are generated deterministically in code.

2 Methodology

2.1 Circuit families and observables

The GHZ family prepares

$$|\mathrm{GHZ}_w\rangle = (|0\rangle^{\otimes w} + |1\rangle^{\otimes w})/\sqrt{2}, \tag{1}$$

with analytic observables $X^{\otimes w} = 1$ and neighboring $ZZ = 1$. The hardware-efficient family alternates fixed-seed single-qubit rotations and nearest-neighbor entanglers at depths 1, 3, and 6. Its Pauli observables and statevector references are fixed before noisy sampling; some references are deliberately away from ± 1 to avoid a stabilizer-only benchmark.

For repeat r, $e_r = \widehat{\mu}_r - \mu_{\mathrm{ref}}$. We report bias, sample variance, and

$$\mathrm{RMSE} = \sqrt{R^{-1}\sum_{r=1}^{R} e_r^2}. \tag{2}$$

Paired RMSE-gain intervals use deterministic bias-corrected and accelerated bootstrap resampling. Decision-error probability is the fraction of repeats whose estimated sign disagrees with the reference decision; Wilson intervals quantify its binomial uncertainty.

2.2 Mitigation and fixed budgets

Raw estimation spends the full budget on the target. Readout mitigation allocates 20% to all 2^w calibration states and 80% to the target. It solves a constrained assignment-matrix correction and reports ill-conditioning explicitly rather than silently inverting an unstable matrix [1]. Linear ZNE splits the same budget over global folds $\lambda \in \{1, 3, 5\}$ and uses the preregistered fit

$$\widehat{\mu}(\lambda) = \beta_0 + \beta_1 \lambda, \qquad \widehat{\mu}_{\mathrm{ZNE}} = \beta_0, \tag{3}$$

without clipping out-of-range extrapolates [3].

2.3 Combined mitigation and factorial interaction

The combined estimator calibrates one full assignment matrix per repeat, corrects the probability distribution at each fold, converts each corrected distribution to an expectation, and then applies the same linear extrapolation. It reserves 20% of the total budget for calibration. The remaining shots are either divided equally across folds (balanced) or allocated 50%/30%/20% to $\lambda = 1, 3, 5$ (base-heavy). Intermediate counts, corrected estimates, extrapolates, and every calibration and folded execution remain in the repeat-level ledger. For paired absolute errors E, the factorial interaction is

$$I = E_{\mathrm{combined}} - E_{\mathrm{readout}} - E_{\mathrm{ZNE}} + E_{\mathrm{raw}}. \tag{4}$$

Negative I indicates that the joint error is below the additive single-method contrast; it does not by itself establish a credible extrapolation mechanism.

2.4 ZNE validity and falsification

An extrapolate is not treated as credible merely because it moves toward the exact reference. A preregistered gate additionally requires retained signal at the largest fold, physical expectation and signed-probability diagnostics, positive paired gain with uncertainty, a positive advantage over matched-cost garbage folding, and no excess of destroyed or unstable repeats. Garbage folding increases execution cost without implementing the intended unitary-equivalent noise scaling and therefore serves only as a negative control. The gate uses 50 deterministic repeats per cell; rejected cells retain every failed criterion and are never converted into ZNE wins.

Table 1: Most simulated operating cells are practical ties; a single universal mitigation winner is not supported.

Cost-aware label	Cells
Practical tie	103
Readout	18
Linear ZNE	14
Inconclusive	9

2.5 Cost-aware labels and frozen selector

For each fully completed cell, the oracle combines RMSE with an execution-overhead penalty. Differences below the preregistered RMSE threshold 0.005 are labeled `practical_tie`; incomplete three-method comparisons are `inconclusive`. A deterministic hash split reserves at least 20% of cells. The selector is then frozen before hold-out outcomes are inspected. It uses only width, depth, declared noise regime, shots, calibration conditioning, and ZNE fit residuals. It can select raw, readout, or ZNE, or abstain; it never receives the statevector reference or oracle label.

3 Experiment Setup

The main grid crosses widths $2, 4, 6$, hardware-efficient depths $1, 3, 6$, noise levels zero/low/medium/high, shot budgets 1,024/4,096/16,384, three methods, and 50 repeats. GHZ depths 3 and 6 remain explicit `not_applicable` cells. All seeds derive from root seed 20261102. Qiskit Aer supplies ideal statevectors and finite-shot noisy simulation [5]. The main model combines independent assignment error with one- and two-qubit depolarizing channels.

A separate 4,096-shot shift set tests intermediate mixed noise, asymmetric readout, coherent over-rotation, and correlated ZZ error. These are controlled synthetic stress tests, not calibrated device models. Every manifest row has a raw row with status `complete`, `failed`, or `not_applicable`; no failed cell is silently dropped.

The combined study preregisters six diagnostic cells: readout-dominated, gate-noise-dominated, practical-tie, ZNE-invalid, 0.02-margin decision, and 0.20-margin decision conditions. It crosses budgets 4,096 and 16,384, both frozen allocation policies, four factorial methods, and 50 paired repeats. Matched-cost garbage folding is used only by the validity gate and is excluded from method ranking.

4 Results

4.1 Operating regimes

Table 1 shows that practical equivalence dominates the grid. Readout and ZNE win distinct subsets, while nine cells are inconclusive because the 50-repeat completeness

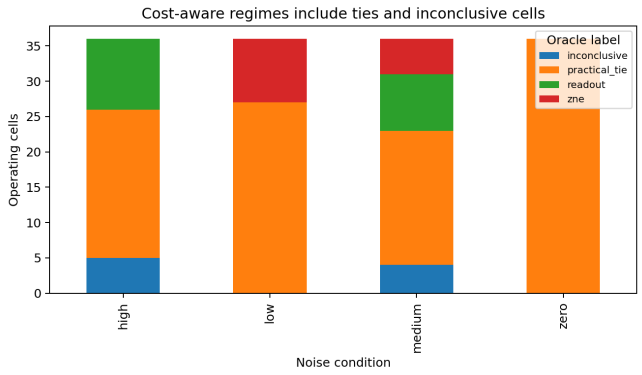


Figure 2: Cost-aware labels retain practical ties and incomplete comparisons instead of forcing every simulated cell to declare a winner.

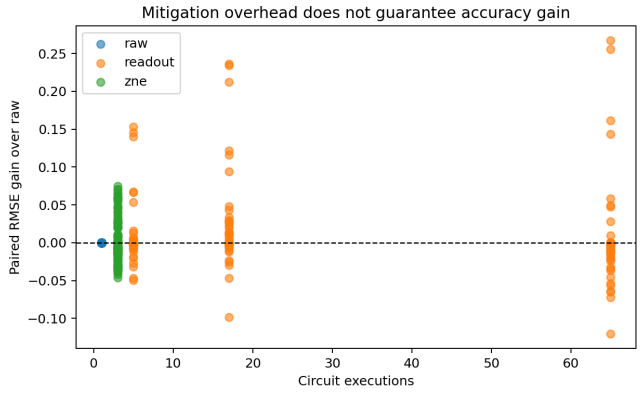


Figure 3: Higher simulated execution overhead does not guarantee positive RMSE gain; negative gains remain visible rather than being clipped or filtered.

contract is not met. Figure 2 shows how these labels vary with simulated noise. The map prevents small numerical RMSE differences from being presented as scientific method changes.

Mitigation can also worsen estimation: 179 of 432 method summaries have negative RMSE gain relative to raw estimation. Readout produces 99 explicit hardware-efficient failures, concentrated at width 6, 1,024 shots, and medium/high noise. These failures are part of method performance, not missing data.

4.2 ZNE validity under negative controls

The probability-level diagnostic contains 800 simulated repeat records across 16 preregistered cells. Of these, 197 repeats are classified as extrapolation-unstable and 5 as signal-destroyed. The cell-level validity gate accepts 5 cells and rejects 11. All 8 hardware-efficient cells are rejected; their apparent corrections do not survive the combined gain-uncertainty and trend requirements. The five accepted cells are Bell-state conditions, so this result narrows rather than broadens the evidence for linear

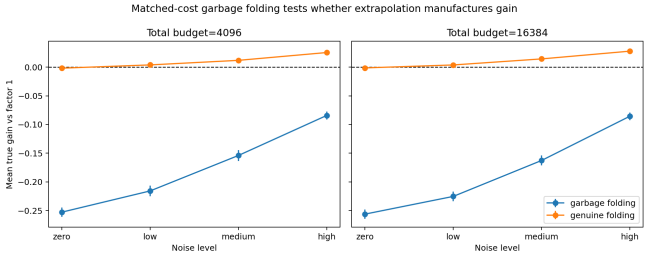


Figure 4: Matched-cost garbage folding separates an apparent correction from a unitary-equivalent ZNE effect; only cells passing the full preregistered gate are reported as credible.

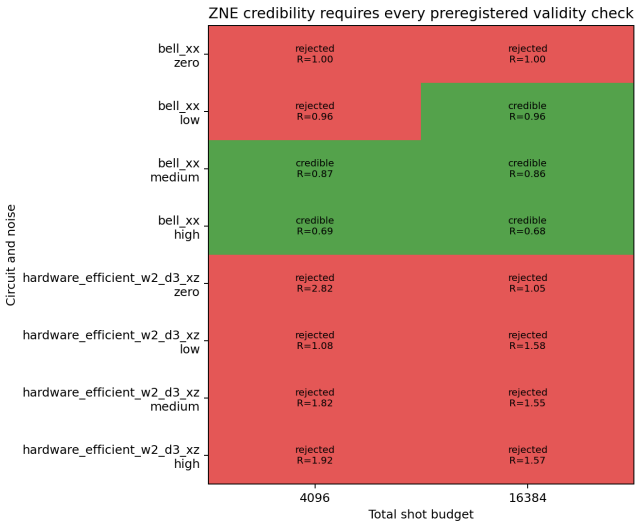


Figure 5: Signal retention and extrapolation stability vary by simulated circuit, noise, and budget; the hardware-efficient cells fail the combined validity gate despite some favorable individual diagnostics.

ZNE.

4.3 Combined QEM under one budget

Combined mitigation has the lowest expectation RMSE in 12 of 16 preregistered circuit–budget–allocation conditions. This descriptive win count is insufficient for a mechanism claim: combined RMSE gain versus raw ranges from negative to positive, and additional execution burden does not guarantee improvement (Fig. 6).

Exactly 12 of 24 paired factorial contrasts are negative; the remainder are positive. Their uncertainty intervals frequently overlap zero (Fig. 7), so the data do not support a uniform cooperative or antagonistic interaction. More decisively, all 16 of 16 combined-ZNE conditions are rejected by at least one frozen signal, physicality, uncertainty, trend, or garbage-control requirement. The RMSE improvements therefore remain simulated estimator outcomes rather than validated ZNE effects.

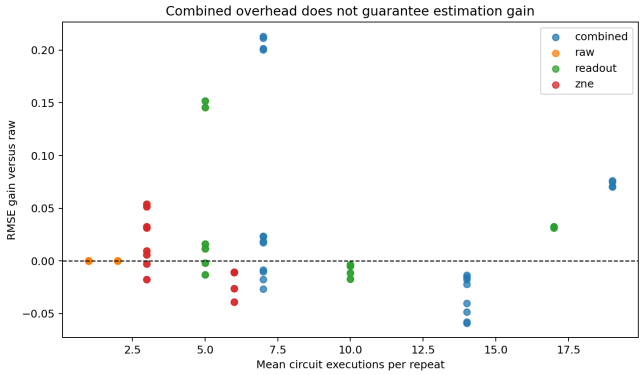


Figure 6: Combined readout correction and ZNE often lowers simulated expectation RMSE, but some conditions remain worse than raw despite greater execution overhead.

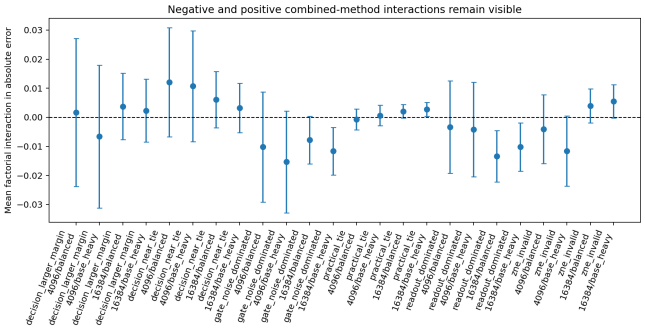


Figure 7: Paired factorial interactions change sign across preregistered cells, budgets, and allocations; many intervals cross zero, rejecting a universal synergy claim.

4.4 Selector holdout and noise shift

On 29 held-out cells, the frozen selector achieves 41.4% decision compatibility. Mean RMSE regret is 0.0207 and median regret is 0.000397. Strict label agreement is lower because a practical tie can contain several acceptable methods; compatibility is therefore the primary decision metric.

Under the four unseen noise shifts, compatibility is 50.0%, mean regret is 0.00771, and median regret is 0. Negative regret can occur because the cost-aware oracle penalizes overhead while the plotted quantity is RMSE alone; it is not evidence that the selector beats the oracle's declared utility.

4.5 Decision reliability

Expectation RMSE and decision reliability are distinct. Figure 10 reports sign-decision error with Wilson uncertainty. A method may reduce continuous estimation error without reducing the probability of a wrong downstream decision.

The controlled combined-QEM decision study sharpens this separation. At margin 0.02, combined decision-error probability ranges from 0.26 to 0.48 across budgets and allocations and does not consistently beat readout alone. At margin 0.20, nearly all methods resolve the decision. Combined mitigation can therefore improve an expectation estimate while leaving a resource-constrained near-tie unresolved.

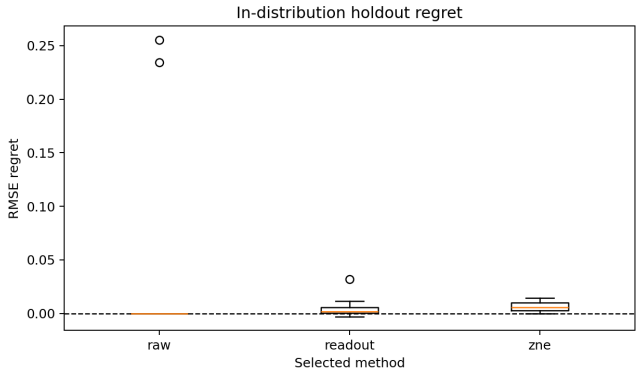


Figure 8: The frozen selector often has small regret but retains substantial positive outliers, so the holdout does not validate a generally reliable policy.

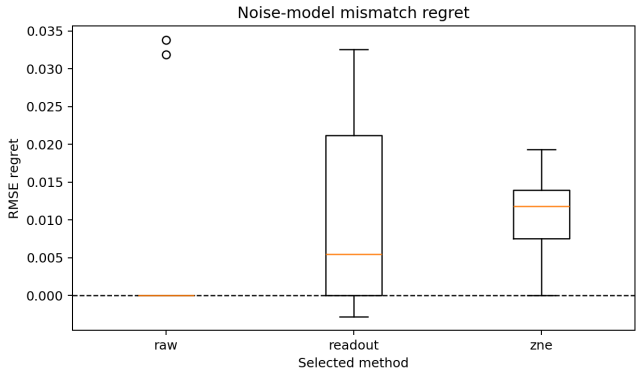


Figure 9: Noise-model mismatch exposes method-dependent regret and outliers; median performance alone would hide these failures.

5 Conclusion

No QEM method is uniformly useful under the declared fixed-budget simulator grid. Most operating cells are practical ties, some favor readout or ZNE, and low-shot readout calibration can fail explicitly. The frozen rule selector is reproducible and explainable, but its modest holdout and shift compatibility make it a baseline for further study rather than a successful adaptive policy. The strongest supported conclusion is operational: mitigation should be compared with raw estimation under equal resource accounting, with ties, negative gains, abstention, and failed ZNE validity checks preserved. In this benchmark, apparent extrapolation gains are more

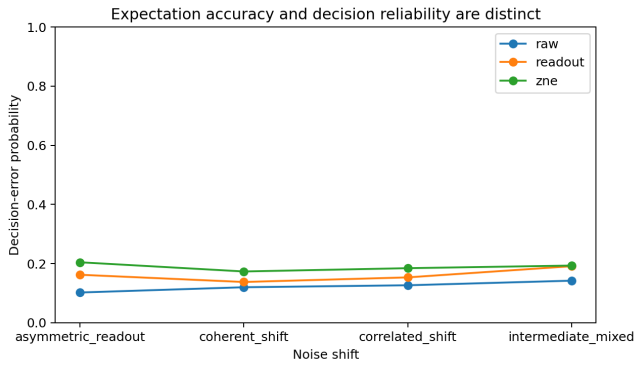


Figure 10: Decision-error probability changes by method and synthetic noise shift, showing that expectation accuracy is not a sufficient proxy for decision reliability.

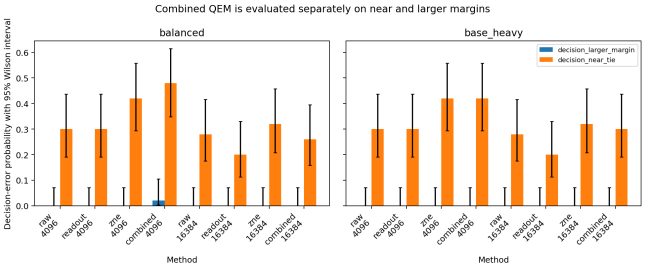


Figure 11: The larger decision margin is resolved, whereas the 0.02 near-tie retains substantial Wilson-bounded error and no consistent combined-QEM advantage.

common than gains that survive the full falsification gate. Combining readout correction with ZNE improves RMSE in selected simulated conditions, but neither its factorial interaction nor its downstream decision error supports a universal synergy claim, and no combined condition passes the frozen validity gate.

6 Discussion

6.1 Supported findings

The data directly support regime dependence, nontrivial execution overhead, explicit readout instability, negative mitigation gains, and a gap between estimation RMSE and decision reliability. The deterministic holdout also shows that intuitive rules do not generalize reliably across the present circuit and noise grid. The ZNE negative control further shows that improvement toward a reference is insufficient evidence unless signal retention, physicality, uncertainty, and a matched-cost control agree. The combined experiment further supports a separation between estimator performance and mechanism validity: lower RMSE can coexist with rejected ZNE diagnostics and unresolved near-tie decisions.

6.2 Hypotheses and limitations

Possible causes of selector error include insufficient features, first-order ZNE, and the mismatch between synthetic training and shift channels; the experiment does not isolate these causes. The validity gate is conditional on its frozen thresholds, linear fit, chosen folds, and maximally mixed observable-floor model; it does not establish a general theorem about ZNE. All noisy values are simulator-derived. There is no IBM or other hardware execution, device drift, calibrated crosstalk, leakage, relaxation, or non-Markovian noise. Width is limited to six, the hardware-efficient ansatz and observables are fixed, and full assignment calibration scales exponentially. The same seed family and simulator implementation bound generalization. No ML selector is trained, and no post-holdout rule tuning is performed. The combined study covers only six preregistered cells, two budgets, and two allocation policies; it does not compare extrapolate-then-correct ordering or establish that readout and gate errors are separable on hardware.

A Reproducibility

```
python -m pytest
python -m src.verify_generated_artifacts
python -m src.generate_paper_assets
tectonic paper/main.tex --outdir paper
```

Repeat-level rows remain under `results/raw`; processed summaries, manifests, run metadata, seeds, diagnostics, and execution counts remain separate and auditable. Tests check analytic references, statevector agreement, folding equivalence, shot accounting, deterministic seeds, manifest coverage, metric recomputation, and the holdout boundary.

References

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